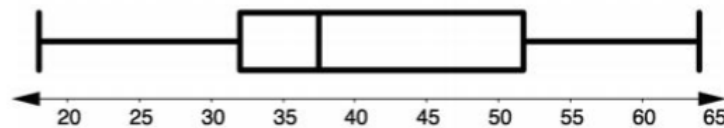


Mr. Ramirez is trying to determine the average score of his student's quiz grades on a recent quiz. He wrote down a summary of the student scores and displayed them in the stem-and-leaf plot and box plot.

Stem	Leaf
1	8, 9
2	5, 8
3	0, 0, 1, 2, 2, 6, 6, 6, 7, 7, 7, 8, 9
4	0, 1, 2, 8
5	1, 2, 5, 8, 8, 9, 9
6	0, 4

Key: 1|8 = 18



1. Which measure of center, the mean or median, best summarizes the quiz scores of the class?

Answers may vary.

Because there aren't any outliers, the mean would best summarize the scores.

2. Find the value of the measure you chose in question 1 if possible.

Mean: $\frac{1,228}{30} \approx 40.9$

Median: 37.5

3. Describe the variation of quiz scores.

Range is $64 - 18 = 46$.

The IQR is $52 - 32 = 20$.

4. Does the range or the interquartile range better represent the variation in quiz scores?

Answers may vary.

Because there aren't any outliers, the range best represents the variation.

Jackie is planning on buying a new car. She has a budget of \$45,000. The prices of new cars in her area are shown below, in thousands of dollars.

23, 52, 42, 38, 56, 70, 40

5. Which measure of center should Jackie use to summarize the prices of the new cars in her area, the mean or the median? Explain your reasoning.
Answers may vary.
Because of the outliers (23 and 70), she should use the median.
6. Which measure of variation should Jackie use to summarize the prices of the new cars in her area, the IQR or the range? Explain your reasoning.
Answers may vary.
IQR because of the outliers.

Jorge likes to listen to music on his phone. The data set below shows the times of the songs on his playlist.

3.25, 1.58, 2.22, 4.03, 3.57, 2.41, 7.46, 3.13, 4.01, 2.59

7. Are there any outliers in the data set? If so, which one?
Explanations may vary.
7.46 is an outlier because it is so much larger than most of the other songs.
8. What measure of center best describes the typical song length on Jorge's playlist? Explain your reasoning.
Explanations may vary.
Median because of the presence of the outlier.
9. What measure of variation best describes the spread of the data distribution? Explain your reasoning.
Explanations may vary.
IQR because of the outlier.
10. Jorge says the IQR is the most accurate depiction of the spread of his song lengths. Do you agree or disagree with Jorge? Explain your reasoning.
Explanations may vary.
I agree because the outlier will affect the range, so the IQR would be more accurate.